

SQL Tables Created:

```
1 CREATE TABLE Hospital (  
2     hospital_name VARCHAR(150) PRIMARY KEY  
3 );  
4  
5 CREATE TABLE Doctor (  
6     doctor_id INT PRIMARY KEY,  
7     doctor_name VARCHAR(100),  
8     hospital_name VARCHAR(150),  
9     CONSTRAINT FK_Doctor_Hospital FOREIGN KEY (hospital_name)  
10    REFERENCES Hospital(hospital_name)  
11 );  
12  
13 CREATE TABLE Billing (  
14     insurance_provider VARCHAR(100) PRIMARY KEY,  
15     billing_amount DECIMAL(10,2)  
16 );  
17  
18 CREATE TABLE Patient (  
19     patient_id INT PRIMARY KEY,  
20     patient_name VARCHAR(100),  
21     age INT,  
22     gender VARCHAR(10),  
23     blood_type CHAR(3),  
24     medical_condition VARCHAR(100),  
25     doctor_id INT,  
26     insurance_provider VARCHAR(100),  
27     CONSTRAINT FK_Patient_Doctor FOREIGN KEY (doctor_id)  
28    REFERENCES Doctor(doctor_id),  
29     CONSTRAINT FK_Patient_Billing FOREIGN KEY  
30    (insurance_provider) REFERENCES Billing(insurance_provider)  
31 );  
32  
33 CREATE TABLE Admission (  
34     admission_id INT IDENTITY(1,1) PRIMARY KEY,  
35     patient_id INT,  
36     admission_date DATE,  
37     discharge_date DATE,  
38     treatment_days INT,  
39     admission_type VARCHAR(20),  
40     room_number INT,  
41     test_results VARCHAR(20),  
42     billing_amount DECIMAL(10,2),  
43     CONSTRAINT FK_Admission_Patient FOREIGN KEY (patient_id)  
44    REFERENCES Patient(patient_id)  
45 );  
46  
47 CREATE TABLE Medical_Record (  
48     medication VARCHAR(100) PRIMARY KEY,  
49     test_results VARCHAR(20)
```

46);

Listing 1: Table Creation

SQL Queries:

Patient Counts:

```
1 -- Query for Patient Count:
2 SELECT COUNT(*) AS total_patients FROM Patient;
```

Gender Counts:

```
1 -- Query for Gender Count:
2 SELECT gender, COUNT(*)
3 FROM Patient
4 GROUP BY gender;
```

Distinct Medical Conditions:

```
1 -- Distinct Medical Conditions:
2 SELECT DISTINCT medical_condition
3 FROM Patient;
```

Patients over 65:

```
1 SELECT medical_condition,
2         COUNT(*) AS number_of_patients_over_65
3 FROM Patient
4 WHERE age > 65
5 GROUP BY medical_condition;
```

Minimum & Maximum Age:

```
1 -- Min/Max age:
2 SELECT MIN(age) AS min_age, MAX(age) AS max_age
3 FROM Patient;
```

Oldest to Youngest Patients:

```

1 --Oldest Patients to youngest:
2 SELECT patient_name,
3         age
4 FROM Patient
5 ORDER BY age DESC;

```

Insurance Check:

```

1 -- Insurance Check
2 SELECT DISTINCT insurance_provider
3 FROM Billing;

```

Top 10 Oldest Patient Information:

```

1 -- Top 10 oldest patient information:
2 SELECT TOP 10 patient_name,
3         medical_condition,
4         age
5 FROM Patient
6 ORDER BY age DESC;

```

Top 10 Youngest Patient Information:

```

1 -- Bottom 10 youngest patient information:
2 SELECT TOP 10 patient_name,
3         medical_condition,
4         age
5 FROM Patient
6 ORDER BY age;

```

Patient Blood Types Grouped:

```

1 -- Blood Types + Patient Count:
2 SELECT blood_type,
3         COUNT(*) as Patient_Count
4 FROM Patient
5 GROUP BY blood_type
6 ORDER BY Patient_Count DESC;

```

Average Age Between Genders:

```

1 SELECT AVG(age) as AverageAge, gender
2 FROM Patient
3 GROUP BY gender
4 ORDER BY AverageAge DESC;

```
